(b) The conditional p.d.f. of Y given $X = 0$ is the ratio of the joint p.f./p.d.f. to $f_1(0)$.

$$g_2(y|0) = \frac{(2y)^0 \exp(-3y)/0!}{(1/3)(2/3)^0} = 3\exp(-3y),$$

for $y > 0$.

(c) The conditional p.d.f. of Y given $X = 1$ is the ratio of the joint p.f./p.d.f. to $f_1(1)$.

$$g_2(y|1) = \frac{(2y)^1 \exp(-3y)/1!}{(1/3)(2/3)^1} = 9y\exp(-3y),$$

for $y > 0$.

(d) The ratio of the two conditional p.d.f.'s is

$$\frac{g_2(y|1)}{g_2(y|0)} = \frac{9y\exp(-3y)}{3\exp(-3y)} = 3y.$$

The ratio is greater than 1 if $y > 1/3$. This corresponds to the intuition that if we observe more calls, then we should think the rate is higher.

13. There are four different treatments on which we are asked to condition. The marginal p.f. of treatment Y is given in the bottom row of Table 3.6 in the text. The conditional p.f. of response given each treatment is the ratio of the two rows above that to the bottom row:

$$g_1(x|1) = \begin{cases} \frac{0.120}{0.267} = 0.4494 & \text{if } x = 0, \\ \frac{0.147}{0.267} = 0.5506 & \text{if } x = 1. \end{cases}$$

$$g_1(x|2) = \begin{cases} \frac{0.087}{0.253} = 0.3439 & \text{if } x = 0, \\ \frac{0.166}{0.253} = 0.6561 & \text{if } x = 1. \end{cases}$$

$$g_1(x|3) = \begin{cases} \frac{0.146}{0.253} = 0.5771 & \text{if } x = 0, \\ \frac{0.107}{0.253} = 0.4229 & \text{if } x = 1. \end{cases}$$

$$g_1(x|4) = \begin{cases} \frac{0.160}{0.227} = 0.7048 & \text{if } x = 0, \\ \frac{0.067}{0.227} = 0.2952 & \text{if } x = 1. \end{cases}$$

The fourth one looks quite different from the others, especially from the second.

3.7 Multivariate Distributions

Commentary

The material around Definition 3.7.8 and Example 3.7.8 reintroduces the concept of conditionally independent random variables. This concept is important in Bayesian inference, but outside of Bayesian inference, it generally appears only in more advanced applications such as expert systems and latent variable models. If an instructor is going to forego all discussion of Bayesian inference then this material (and Exercises 13 and 14) could be skipped.

Solutions to Exercises

1. (a) We have

$$\int_0^1 \int_0^1 \int_0^1 f(x_1, d_2, x_3)\, dx_1\, dx_2\, dx_3 = 3c.$$

Since the value of this integral must be equal to 1, it follows that $c = 1/3$.

(b) For $0 \le x_1 \le 1$ and $0 \le x_3 \le 1$,

$$f_{13}(x_1, x_3) = \int_0^1 f(x_1, x_2, x_3)\, dx_2 = \frac{1}{3}(x_1 + 1 + 3x_3).$$

(c) The conditional p.d.f. of x_3 given that $x_1 = 1/4$ and $x_2 = 3/4$ is, for $0 \le x_3 \le 1$,

$$g_3\left(x_3 \,\middle|\, x_1 = \frac{1}{4}, x_2 = \frac{3}{4}\right) = \frac{f\left(\frac{1}{4}, \frac{3}{4}, x_3\right)}{f_{12}\left(\frac{1}{4}, \frac{3}{4}\right)} = \frac{7}{13} + \frac{12}{13}x_3.$$

Therefore,

$$\Pr\left(X_3 < \frac{1}{2} \,\middle|\, X_1 = \frac{1}{4}, X_2 = \frac{3}{4}\right) = \int_0^{\frac{1}{2}} \left(\frac{7}{13} + \frac{12}{13}x_3\right) dx_3 = \frac{5}{13}.$$

2. (a) First, integrate over x_1. We need to compute $\int_0^1 c x_1^{1+x_2+x_3}(1-x_1)^{3-x_2-x_3} dx_1$. The two exponents always add to 4 and each is always at least 1. So the possible pairs of exponents are $(1,3)$, $(2,2)$, and $(3,1)$. By the symmetry of the function, the first and last will give the same value of the integral. In this case, the values are

$$\int_0^1 c[x_1^3 - x_1^4] dx_1 = \frac{c}{4} - \frac{c}{5} = \frac{c}{20}. \tag{S.3.2}$$

In the other case, the integral is

$$\int_0^1 c[x_1^2 - 2x_1^3 + x_1^4] dx_1 = \frac{c}{3} - \frac{2c}{4} + \frac{c}{5} = \frac{c}{30}. \tag{S.3.3}$$

Finally, sum over the possible (x_2, x_3) pairs. The mapping between (x_2, x_3) values and the exponents in the integral is as follows:

(x_2, x_3)	Exponents
$(0,0)$	$(1,3)$
$(0,1)$	$(2,2)$
$(1,0)$	$(2,2)$
$(1,1)$	$(3,1)$

Summing over the four possible (x_2, x_3) pairs gives the sum of $c/6$, so $c = 6$.

(b) The marginal joint p.f. of (X_2, X_3) is given by setting $c = 6$ in (S.3.2) and (S.3.3) and using the above table.

$$f_{23}(x_2, x_3) = \begin{cases} 0.3 & \text{if } (x_2, x_3) \in \{(0,0), (1,1)\}, \\ 0.2 & \text{if } (x_2, x_3) \in \{(1,0), (0,1)\}. \end{cases}$$

(c) The conditional p.d.f. of X_1 given $X_2 = 1$ and $X_3 = 1$ is $1/0.3$ times the joint p.f./p.d.f. evaluated at $x_2 = x_3 = 1$:

$$g_1(x_1|1,1) = \begin{cases} 20x_1^3(1-x_1) & \text{if } 0 < x_1 < 1, \\ 0 & \text{otherwise.} \end{cases}$$

3. The p.d.f. should be positive for all $x_i > 0$ not just for all $x_i > 1$ as stated in early printings. This will match the answers in the back of the text.

(a) We have

$$\int_0^\infty \int_0^\infty \int_0^\infty f(x_1, x_2, x_3)\, dx_1\, dx_2\, dx_3 = \frac{1}{6}c.$$

Since the value of this integral must be equal to 1, it follows that c = 6. If one used $x_i > 1$ instead, then the integral would equal $\exp(-6)/6$, so that $c = 6\exp(6)$.

(b) For $x_1 > 0$, $x_3 > 0$,

$$f_{13}(x_1, x_3) \int_0^\infty f(x_1, x_2, x_3)\, dx_2 = 3\exp[-(x_1 + 3x_3)].$$

If one used $x_i > 1$ instead, then for $x_1 > 1$ and $x_3 > 1$, $f_{13}(x_1, x_3) = 3\exp(-x_1 - 3x_3 + 4)$.

(c) It is helpful at this stage to recognize that the random variables X_1, X_2, and X_3 are independent because their joint p.d.f. $f(x_1, x_2, x_3)$ can be factored as in Eq. (3.7.7); i.e., for $x_i > 0$ $(i = 1, 2, 3)$,

$$f(x_1, x_2, x_3) = (\exp(-x_1)) = (2\exp(-x_2))(3\exp(-x_3)).$$

It follows that

$$\Pr(X_1 < 1 \,|\, X_2 = 2, X_3 = 1) = Pr(X_1 < 1) = \int_0^1 f_1(x_1)\, dx_1 = \int_0^1 \exp(-x_1) dx_1 = 1 - \frac{1}{e}.$$

This answer could also be obtained without explicitly using the independence of X_1, X_2, and X_3 by calculating first the marginal joint p.d.f.

$$f_{23}(x_2, x_3) = \int_0^\infty f(x_1, x_2, x_3)\, dx_1,$$

then calculating the conditional p.d.f.

$$g_1(x_1 \,|\, x_2 = 2, x_3 = 1) = \frac{f(x_1, 2, 1)}{f_{2,3}(2, 1)},$$

and finally calculating the probability

$$\Pr(X_1 < 1 \,|\, X_2 = 2, X_3 = 1) = \int_0^1 g_1(x_1 \,|\, x_2 = 2, x_3 = 1) dx_1.$$

If one used $x_i > 1$ instead, then the probability in this part is 0.

4. The joint p.d.f. $f(x_1, x_2, x_3)$ is constant over the cube S. Since

$$\iiint_S dx_1\, dx_2\, dx_3 = \int_0^1 \int_0^1 \int_0^1 dx_1\, dx_2\, dx_3 = 1,$$

it follows that $f(x_1, x_2, x_3) = 1$ for $(x_1, x_2, x_3) \in S$. Hence, the probability of any subset of S will be equal to the volume of that subset.

(a) The set of points such that $(x_1 - 1/2)^2 + (x_2 - 1/2)^2 + (x_3 - 1/2)^2 \leq 1/4$ is a sphere of radius 1/2 with center at the point $(1/2, 1/2, 1/2,)$. Hence, this sphere is entirely contained with in the cube S. Since the volume of any sphere is $4\pi r^3/3$, the volume of this sphere, and also its probability, is $4\pi(1/2)^3/3 = \pi/6$.

(b) The set of points such that $c_1^2 + x_2^2 + x_3^2 \leq 1$ is a sphere of radius 1 with center at the origin (0, 0, 0). Hence, the volume of this sphere is $4\pi/3$. However, only one octant of this sphere, the octant in which all three coordinates are nonnegative, lies in S. Hence, the volume of the intersection of the sphere with the set S, and also its probability, is $\frac{1}{8} \cdot \frac{4}{3}\pi = \frac{1}{6}\pi$.

5. (a) The probability that all n independent components will function properly is the product of their individual probabilities and is therefore equal to $\prod_{i=1}^{n} p_i$.

   (b) The probability that all n independent components will not function properly is the product of their individual probabilities of not functioning properly and is therefore equal to $\prod_{i=1}^{n}(1-p_i)$. The probability that at least one component will function properly is $1-\prod_{i=1}^{n}(1-p_i)$.

6. Since the n random variables $x_1, \ldots, x_n$ are i.i.d. and each has the p.f. f, the probability that a particular variable X_i will be equal to a particular value x is $f(x)$, and the probability that all n variables will be equal to a particular value x is $[f(x)]^n$. Hence, the probability that all n variables will be equal, without any specification of their common value, is $\sum_x [f(x)]^n$.

7. The probability that a particular variable X_i will lie in the interval (a, b) is $p = \int_a^b f(x)\,dx$. Since the variables $X_1, \ldots, X_n$ are independent, the probability that exactly i of these variables will lie in the interval (a, b) is $\binom{n}{i} p^i (1-p)^{n-i}$. Therefore, the required probability is

$$\sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}.$$

8. For any given value x of X, the random variables $Y_1, \ldots, Y_n$ are i.i.d., each with the p.d.f. $g(y \mid x)$. Therefore, the conditional joint p.d.f. of $Y_1 \ldots, Y_n$ given that $X = x$ is

$$h(y_1, \ldots, y_n \mid x) = g(y_1 \mid x) \ldots g(y_n \mid x) = \begin{cases} \dfrac{1}{x^n} & \text{for } 0 < y_i < x,\ i = 1, \ldots, n, \\ 0 & \text{otherwise.} \end{cases}$$

   The joint p.d.f. of X and $Y_1, \ldots, Y_n$ is, therefore,

$$f(x)h(y_1, \ldots, y_n \mid x) = \begin{cases} \dfrac{1}{n!}\exp(-x) & \text{for } \ 0 < y_i < x \quad (i = 1, \ldots, n), \\ 0 & \text{otherwise.} \end{cases}$$

   This joint p.d.f. is positive if and only if each $y_i > 0$ and x is greater than every y_i. In other words, x must be greater than $m = \max\{y_1, \ldots, y_n\}$.

   (a) For $y_1 > 0$ $(i = 1, \ldots, n)$, the marginal joint p.d.f. of $Y_1, \ldots, Y_n$ is

$$g_0(y_1, \ldots, y_n) = \int_{-\infty}^{\infty} f(x)h(y_1, \ldots, y_n \mid x)\,dx = \int_m^{\infty} \frac{1}{n!}\exp(-x)\,dx = \frac{1}{n!}\exp(-m).$$

   (b) For $y_i > 0$ $(i = 1, \ldots, n)$, the conditional p.d.f. of X given that $Y_i = y_i (i = 1, \ldots, n)$ is

$$g_1(x \mid y_1, \ldots, y_n) = \frac{f(x)h(y_1, \ldots, y_n \mid x)}{g_0(y_1, \ldots, y_n)} = \begin{cases} \exp(-(x-m)) & \text{for } x > m, \\ 0 & \text{otherwise.} \end{cases}$$

9. (a) Since $X_i = X$ for $i = 1, 2$, we know that X_i has the same distribution as X. Since X has a continuous distribution, then so does X_i for $i = 1, 2$.

   (b) We know that $\Pr(X_1 = X_2) = 1$. Let $A = \{(x_1, x_2) : x_1 = x_2\}$. Then $\Pr((X_1, X_2) \in A) = 1$. However, for every function f, $\int_A \int f(x_1, x_2) dx_1 dx_2 = 0$. So there is no possible joint p.d.f.

10. The marginal p.d.f. of Z is $2\exp(-2z)$, for $z > 0$. The coordinates of $\boldsymbol{X}$ are conditionally i.i.d. given $Z = z$ with p.d.f. $z\exp(-zx)$, for $x > 0$. This makes the joint p.d.f. of $(Z, \boldsymbol{X})$ equal to $2z^5\exp(-z[2 + x_1 + \cdots + x_5])$ for all variables positive. The marginal joint p.d.f. of $\boldsymbol{X}$ is obtained by integrating z out of this.

$$f_2(\boldsymbol{x}) = \int_0^\infty 2z^5\exp(-z[2 + x_1 + \cdots + x_5])dz = \frac{240}{(2 + x_1 + \cdots + x_5)^6}, \text{ for all } x_i > 0.$$

Here, we use the formula $\int_0^\infty y^k\exp(-y)dy = k!$ from Exercise 12 in Sec. 3.6. The conditional p.d.f. of Z given $\boldsymbol{X} = (x_1, \ldots, x_5)$ is then

$$g_1(z|\boldsymbol{x}) = \frac{(2 + x_1 + \cdots + x_5)^6}{120}z^5\exp(-z[2 + x_1 + \cdots + x_5]),$$

for $z > 0$.

11. Since $X_1, \ldots, X_n$ are independent, their joint p.f., p.d.f., or p.f./p.d.f. factors as

$$f(x_1, \ldots, x_n) = f_1(x_1)\cdots f_n(x_n),$$

where each f_i is a p.f. or p.d.f. If we sum or integrate over all x_j such that $j \notin \{i_1, \ldots, i_k\}$ we obtain the joint p.f., p.d.f., or p.f./p.d.f. of $X_{i_1}, \ldots, X_{i_k}$ equal to $f_{i_1}(x_{i_1})\cdots f_{i_k}(x_{i_k})$, which is factored in a way that makes it clear that $X_{i_1}, \ldots, X_{i_k}$ are independent.

12. Let $h(\boldsymbol{y}, \boldsymbol{w})$ be the marginal joint p.d.f. of $\boldsymbol{Y}$ and $\boldsymbol{W}$, and let $h_2(\boldsymbol{w})$ be the marginal p.d.f. of $\boldsymbol{w}$. Then

$$\begin{aligned}
h(\boldsymbol{y}, \boldsymbol{w}) &= \int f(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{w})d\boldsymbol{z}, \\
h_2(\boldsymbol{w}) &= \int\int f(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{w})d\boldsymbol{z}d\boldsymbol{y}, \\
g_1(\boldsymbol{y}, \boldsymbol{z}|\boldsymbol{w}) &= \frac{f(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{w})}{h_2(\boldsymbol{w})}, \\
g_2(\boldsymbol{y}|\boldsymbol{w}) &= \frac{h(\boldsymbol{y}, \boldsymbol{w})}{h_2(\boldsymbol{w})} = \frac{\int f(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{w})d\boldsymbol{z}}{h_2(\boldsymbol{w})} = \int g_1(\boldsymbol{y}, \boldsymbol{z}|\boldsymbol{w})d\boldsymbol{z}.
\end{aligned}$$

13. Let $f(x_1, x_2, x_3, z)$ be the joint p.d.f. of (X_1, X_2, X_3, Z). Let $f_{12}(x_1, x_2)$ be the marginal joint p.d.f. of (X_1, X_2). The the conditional p.d.f. of X_3 given $(X_1, X_2) = (x_1, x_2)$ is

$$\frac{\int f(x_1, x_2, x_3, z)dz}{f_{12}(x_1, x_2)} = \frac{\int g(x_1|z)g(x_2|z)g(x_3|z)f_0(z)dz}{f_{12}(x_1, x_2)} = \int g(x_3|z)\frac{g(x_1|z)g(x_2|z)f_0(z)}{f_{12}(x_1, x_2)}dz.$$

According to Bayes' theorem for random variables, the fraction in this last integral is $g_0(z|x_1, x_2)$. Using the specific formulas in the text, we can calculate the last integral as

$$\begin{aligned}
&\int_0^\infty z\exp(-zx_3)\frac{1}{2}(2 + x_1 + x_2)^3z^2\exp(-z(2 + x_1 + x_2))dx \\
&= \frac{(2 + x_1 + x_2)^3}{2}\int_0^\infty z^3\exp(-z(2 + x_1 + x_2 + x_3))dz \\
&= \frac{(2 + x_1 + x_2)^3}{2}\frac{6}{(2 + x_1 + x_2 + x_3)^4} = \frac{3(2 + x_1 + x_2)^3}{(2 + x_1 + x_2 + x_3)^4}.
\end{aligned}$$

The joint p.d.f. of (X_1, X_2, X_3) can be computed in a manner similar to the joint p.d.f. of (X_1, X_2) and it is

$$f_{123}(x_1, x_2, x_3) = \frac{12}{(2 + x_1 + x_2 + x_3)^4}.$$

The ratio of $f_{123}(x_1, x_2, x_3)$ to $f_{12}(x_1, x_2)$ is the conditional p.d.f. calculated above.

14. (a) We can substitute $x_1 = 5$ and $x_2 = 7$ in the conditional p.d.f. computed in Exercise 13.

$$g_3(x_3|5, 7) = \frac{3(2 + 5 + 7)^3}{(2 + 5 + 7 + x_3)^4} = \frac{8232}{(14 + x_3)^4},$$

for $x_3 > 0$.

(b) The conditional probability we want is the integral of the p.d.f. above from 3 to ∞.

$$\int_3^\infty \frac{8232}{(14 + x_3)^4} dx_3 = -\frac{2744}{(14 + x_3)^3}\Big|_{x_3=3}^\infty = 0.5585.$$

In Example 3.7.9, we computed the marginal probability $\Pr(X_3 > 3) = 0.4$. Now that we have observed two service times that are both longer than 3, namely 5 and 7, we think that the probability of $X_3 > 3$ should be larger.

15. Let A be an arbitrary n-dimensional set. Because $\Pr(W = c) = 1$, we have

$$\Pr((X_1, \ldots, X_n) \in A, W = w) = \begin{cases} \Pr(X_1, \ldots, X_n) \in A) & \text{if } w = c, \\ 0 & \text{otherwise.} \end{cases}$$

It follows that

$$\Pr((X_1, \ldots, X_n) \in A|W = w) = \begin{cases} \Pr(X_1, \ldots, X_n) \in A) & \text{if } w = c, \\ 0 & \text{otherwise.} \end{cases}$$

Hence the conditional joint distribution of $X_1, \ldots, X_n$ given W is the same as the unconditional joint distribution of $X_1, \ldots, X_n$, which is the distribution of independent random variables.

3.8 Functions of a Random Variable

Commentary

A brief discussion of simulation appears at the end of this section. This can be considered a teaser for the more detailed treatment in Chapter 12. Simulation is becoming a very important tool in statistics and applied probability. Even those instructors who prefer not to cover Chapter 12 have the option of introducing the topic here for the benefit of students who will need to study simulation in more detail in another course.

If you wish to use the statistical software *R*, then the function `runif` will be most useful. For the purposes of this section, `runif(n)` will return `n` pseudo-uniform random numbers on the interval $[0, 1]$. Of course, either `n` must be assigned a value before expecting *R* to understand `runif(n)`, or one must put an explicit value of `n` into the function. The following two options both produce 1,000 pseudo-uniform random numbers and store them in an object called `unumbs`:

- `unumbs=runif(1000)`
- `n=1000`
  `unumbs=runif(n)`